



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-2

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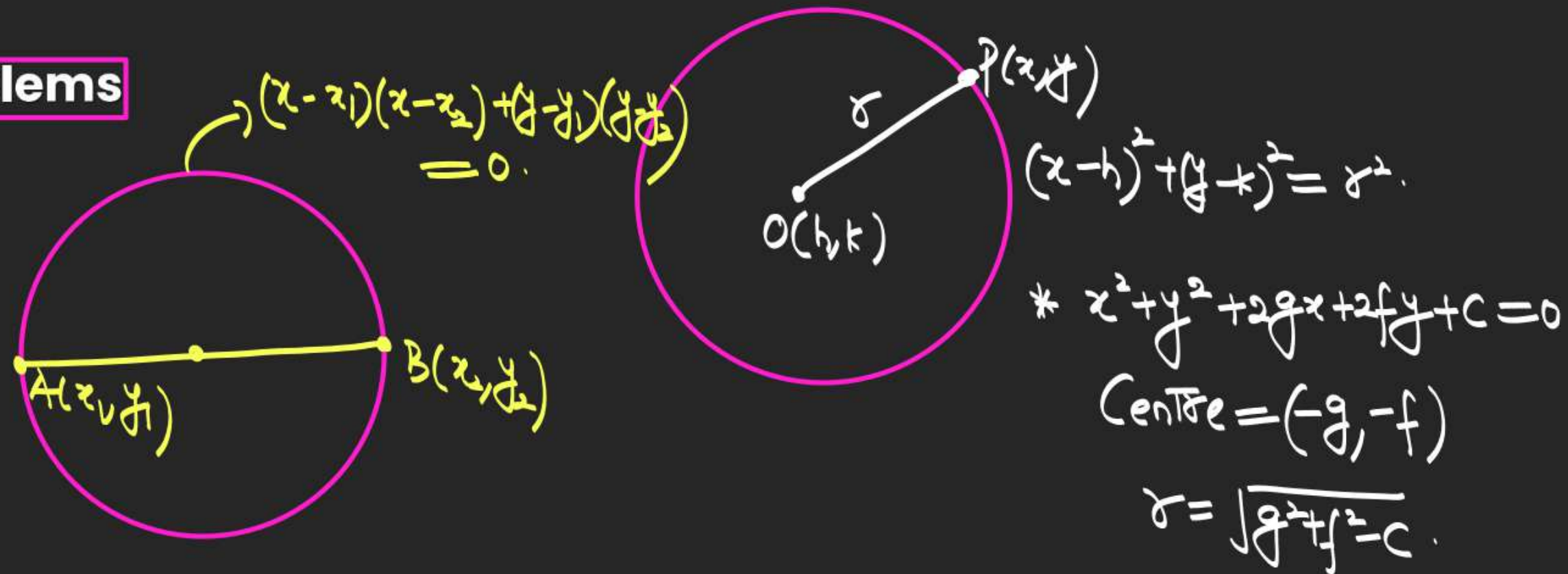
TOPICS TO BE COVERED

1

Exercise Discussion 10.1

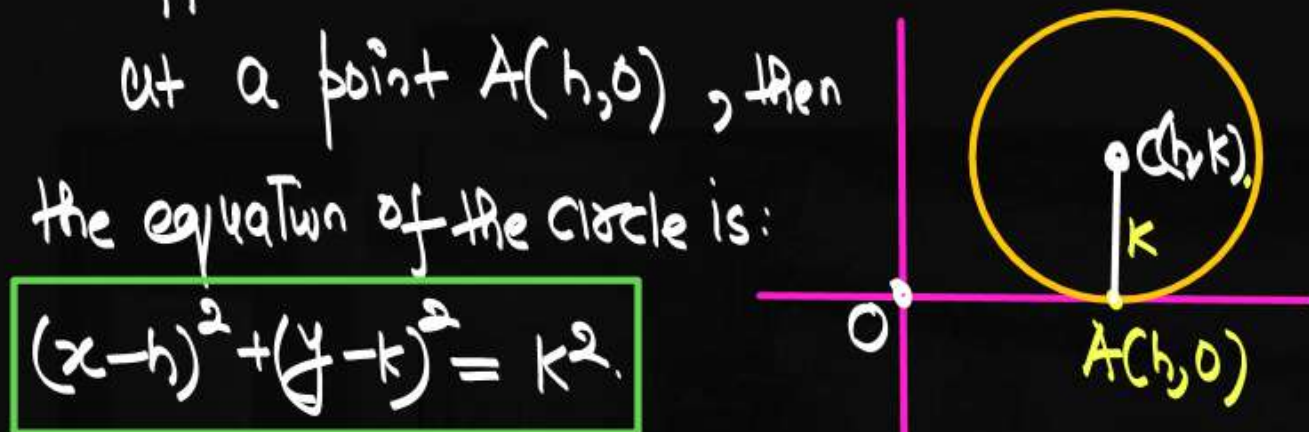
2

Related Problems



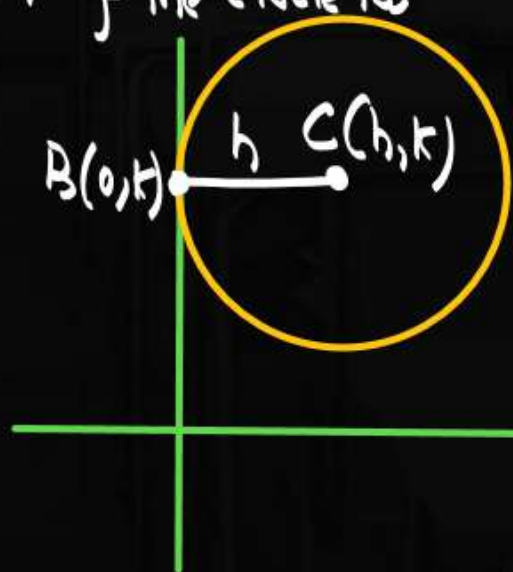
Equation of Circles Touching the Co-ordinate axis:

* Suppose that the circle touches the x -axis



* Suppose that the circle touches the y -axis at a point $B(0, k)$, then the equation of the circle is:

$$(x-h)^2 + (y-k)^2 = h^2$$

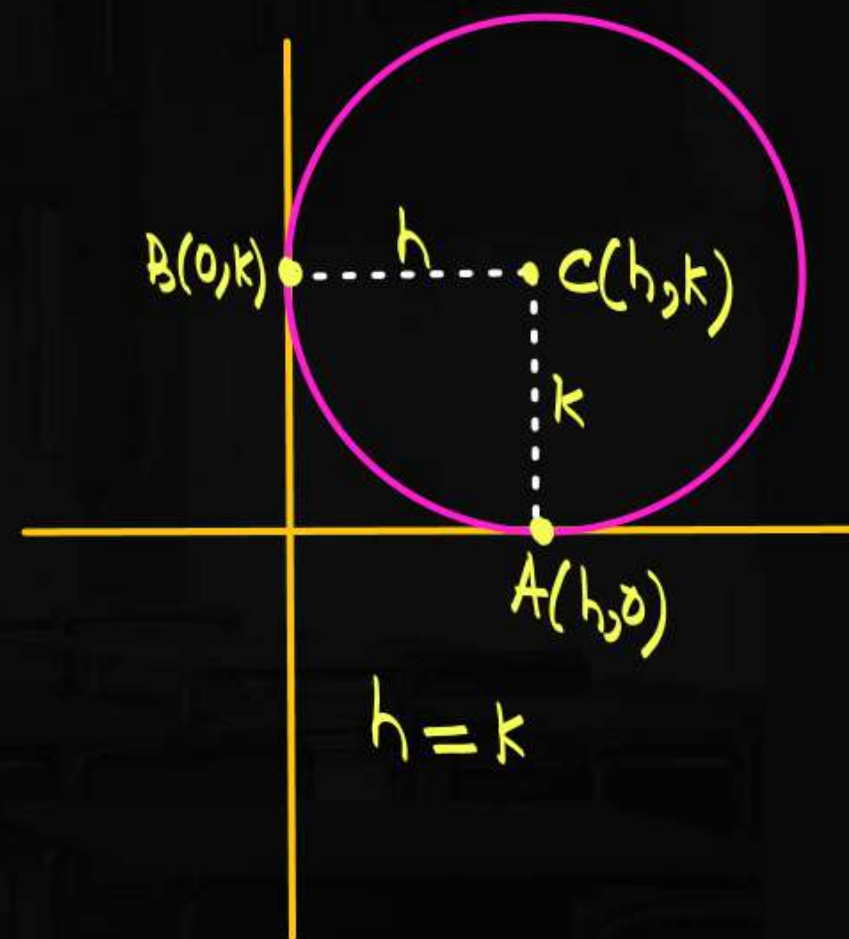


* Suppose that the circle touches the x -axis at $A(h, 0)$ & y -axis at $B(0, k)$, then the equation of the circle is;

$$(x-h)^2 + (y-k)^2 = h^2$$

or

$$(x-k)^2 + (y-k)^2 = k^2$$



QUESTION (NCERT EXAMPLER):

The equation of the circle in the first quadrant touching each coordinate axis at a distance of one unit from the origin is:

1 ✓ $x^2 + y^2 - 2x - 2y + 1 = 0$

2 ✗ $x^2 + y^2 - 2x - 2y - 1 = 0$

3 ✗ $x^2 + y^2 - 2x - 2y = 0$

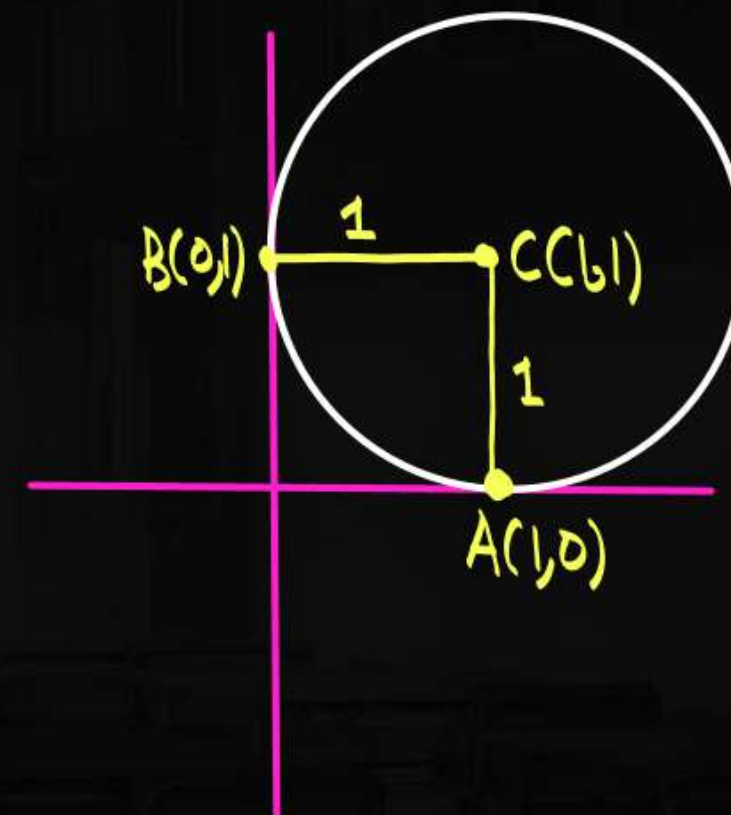
4 ✗ $x^2 + y^2 - 2x + 2y - 1 = 0$

(Sol) Required equation is:

$$(x-1)^2 + (y-1)^2 = 1^2$$

$$\Rightarrow x^2 - 2x + 1 + y^2 - 2y + 1 = 1$$

$$\Rightarrow x^2 + y^2 - 2x - 2y + 1 = 0$$



QUESTION (NCERT EXAMPLER):

The circle passing through $(1, -2)$ and touching the axis of x at $(3, 0)$ also passes through the point

(Homework Problem)

1 $(-5, 2)$

2 $(2, -5)$

3 $(5, -2)$

4 $(-2, 5)$

QUESTION (NCERT EXAMPLER):

The equation of the circle having centre $(1, -2)$ and passing through the point of intersection of the lines $3x + y = 14$ and $2x + 5y = 18$ is.

HP/W

1 $x^2 + y^2 - 2x + 4y - 20 = 0$

2 $x^2 + y^2 - 2x - 4y - 20 = 0$

3 $x^2 + y^2 + 2x - 4y - 20 = 0$

4 $x^2 + y^2 + 2x + 4y - 20 = 0$

QUESTION (NCERT EXAMPLER):

Equation of a circle which passes through (3, 6) and touches the axes is.

Home work.
Problem.

1 $x^2 + y^2 + 6x + 6y + 3 = 0$

2 $x^2 + y^2 - 6x - 6y - 9 = 0$

3 $x^2 + y^2 - 6x - 6y + 9 = 0$

4 None of these

QUESTION (NCERT EXAMPLER):

Equation of the circle with centre on the y -axis and passing through the origin and the point $(2, 3)$ is.

1 ☒ $x^2 + y^2 + 6x + 6y + 3 = 0$

2 ☒ $3x^2 + 3y^2 + 13y = 0$

3 ☒ $3x^2 + 3y^2 - 13y = 0$

4 ☒ None of these

(Sol) Required equation is:

$$x^2 + (y - k)^2 = k^2 \quad \text{--- (1)}$$

Since (1) passes through $(2, 3)$. So

$$2^2 + (3 - k)^2 = k^2$$

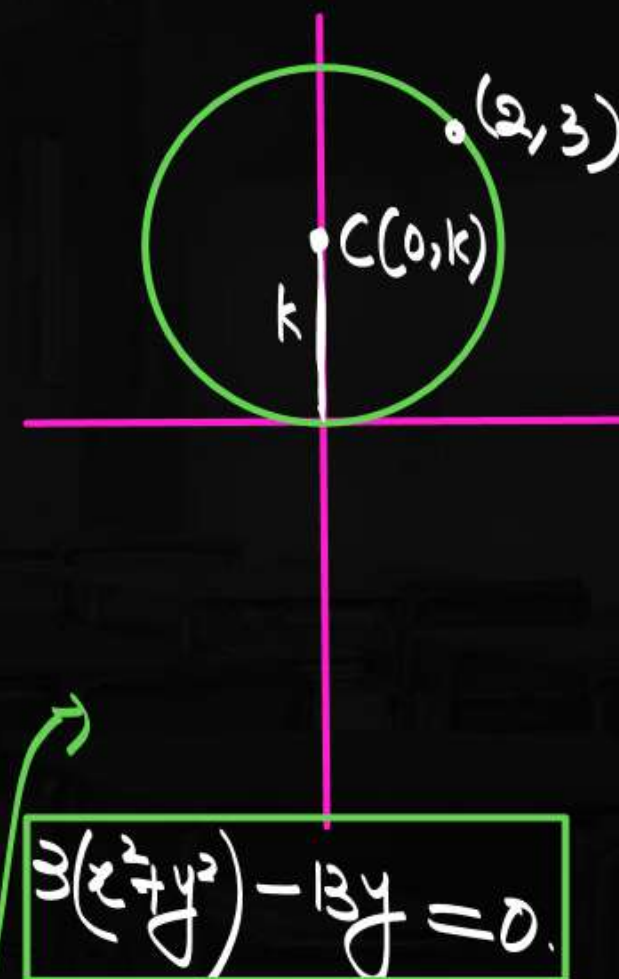
$$\Rightarrow 4 + 9 + k^2 - 6k = k^2$$

$$\Rightarrow 13 - 6k = 0$$

$$\Rightarrow k = \frac{13}{6}$$

So from (1), we get:

$$x^2 + y^2 + \left(\frac{13}{6}\right)^2 - 2 \times \frac{13}{6} y = \left(\frac{13}{6}\right)^2$$



QUESTION (NCERT EXAMPLER):

Centroid divides the median in 2:1 ratio.

The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is

1 ☒ $x^2 + y^2 = 9a^2$

2 ☒ $x^2 + y^2 = 16a^2$

3 ☒ $x^2 + y^2 = 4a^2$

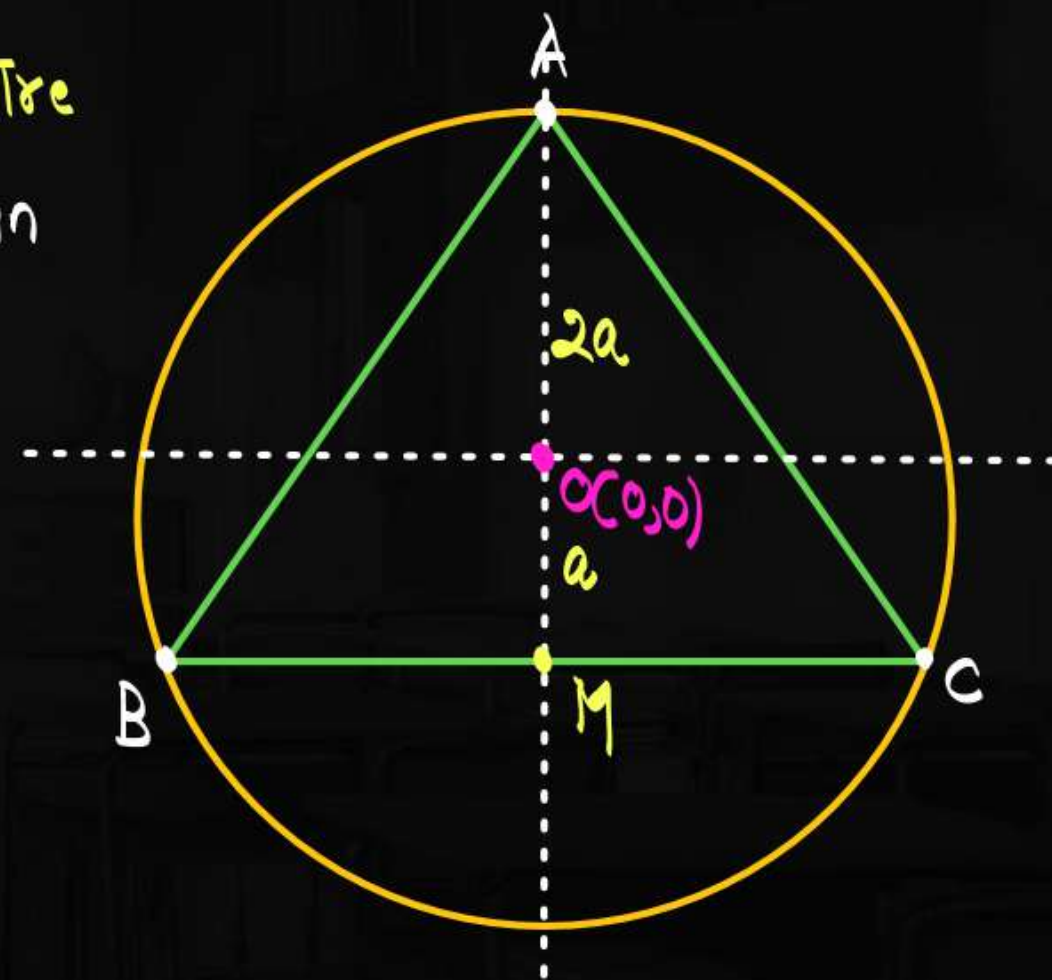
4 ☒ $x^2 + y^2 = a^2$

(Sol) We claim that $O(0,0)$ is the circumcentre or orthocentre or centroid or incentre of an equilateral triangle $\triangle ABC$ (check it)

Required equation is:

$$(x-0)^2 + (y-0)^2 = (2a)^2$$

$$\Rightarrow x^2 + y^2 = 4a^2$$



EXERCISE – 10.1

QUESTION

In each of the following Exercises 1 to 5, find the equation of the circle with
Centre $(0, 2)$ and radius 2

$$(x-h)^2 + (y-k)^2 = r^2.$$

$$(x-0)^2 + (y-2)^2 = 2^2$$

$$\Rightarrow x^2 + y^2 - 4y = 4$$

$$\Rightarrow x^2 + y^2 - 4y = 0.$$

QUESTION

In each of the following Exercises 1 to 5, find the equation of the circle with
Centre $(-2, 3)$ and radius 4

H/W

QUESTION

In each of the following Exercises 1 to 5, find the equation of the circle with

Centre $\left(\frac{1}{2}, \frac{1}{4}\right)$ and radius $\frac{1}{12}$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{4}\right)^2 = \left(\frac{1}{12}\right)^2$$

H/W

QUESTION

In each of the following Exercises 1 to 5, find the equation of the circle with
Centre $(1, 1)$ and radius $\sqrt{2}$

$$(x-1)^2 + (y-1)^2 = \sqrt{2}^2$$

$$\Rightarrow x^2 - 2x + 1 + y^2 + 1 - 2y = 2$$

$$\Rightarrow x^2 + y^2 - 2x - 2y + \cancel{2} = \cancel{2}$$

$$\Rightarrow x^2 + y^2 - 2x - 2y = 0$$

QUESTION

In each of the following Exercises 1 to 5, find the equation of the circle with

Centre $(-a, -b)$ and radius $\sqrt{a^2 - b^2}$

h/w

$$(x+a)^2 + (y+b)^2 = \cancel{\sqrt{a^2 - b^2}}$$

QUESTION

In each of the following Exercises 6 to 9, find the centre and radius of the circles.

$$(x + 5)^2 + (x - 3)^2 = 36$$

$$(h, k) = (-5, 3)$$

$$r = 6$$

QUESTION

In each of the following Exercises 6 to 9, find the centre and radius of the circles.

$$x^2 + y^2 - 4x - 8y - 45 = 0$$

(Sol) Here $2g = -4$, $2f = -8$, $c = -45$.

$$\Rightarrow g = -2, f = -4, c = -45.$$

So Centre $= (-g, -f) = (2, 4)$

$$\text{Radius, } r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 16 + 45}$$

$$\Rightarrow r = \sqrt{65} \text{ units.}$$

QUESTION

In each of the following Exercises 6 to 9, find the centre and radius of the circles.

$$x^2 + y^2 - 8x - 10y - 12 = 0$$

h/w

QUESTION

In each of the following Exercises 6 to 9, find the centre and radius of the circles.

$$2x^2 + 2y^2 - x = 0$$

$$2(x^2 + y^2) - x = 0$$

$$x^2 + y^2 - \frac{1}{2}x = 0$$

$$\left(\frac{1}{4}, 0\right) \quad r = \sqrt{\left(\frac{1}{2}\right)^2 + 0^2 - 0}$$

$$r = \frac{1}{2}$$

QUESTION

Find the equation of the circle passing through the points $(4, 1)$ and $(6, 5)$ and whose centre is on the line $4x + y = 16$.

(Sol) Suppose that the required circle be:

$$(x-h)^2 + (y-k)^2 = r^2 \quad \text{--- (1)}$$

Since (1) passes through $(4, 1)$ & $(6, 5)$, so,

$$(4-h)^2 + (1-k)^2 = r^2 \quad \text{--- (2)}$$

$$\& (6-h)^2 + (5-k)^2 = r^2 \quad \text{--- (3)}$$

From (2) & (3), we get,

$$(4-h)^2 + (1-k)^2 = (6-h)^2 + (5-k)^2$$

$$\Rightarrow 16 + \cancel{h^2} - 8h + 1 + \cancel{k^2} - 2k = 36 + \cancel{h^2} - 12h + 25 + \cancel{k^2} - 10k$$

$$\Rightarrow 17 - 8h - 2k = 61 - 12h - 10k$$

$$\Rightarrow -8h - 2k + 12h + 10k = 61 - 17$$

$$\Rightarrow 4h + 8k = 44$$

$$\Rightarrow \cancel{4}(h+2k) = \cancel{4} \times 11$$

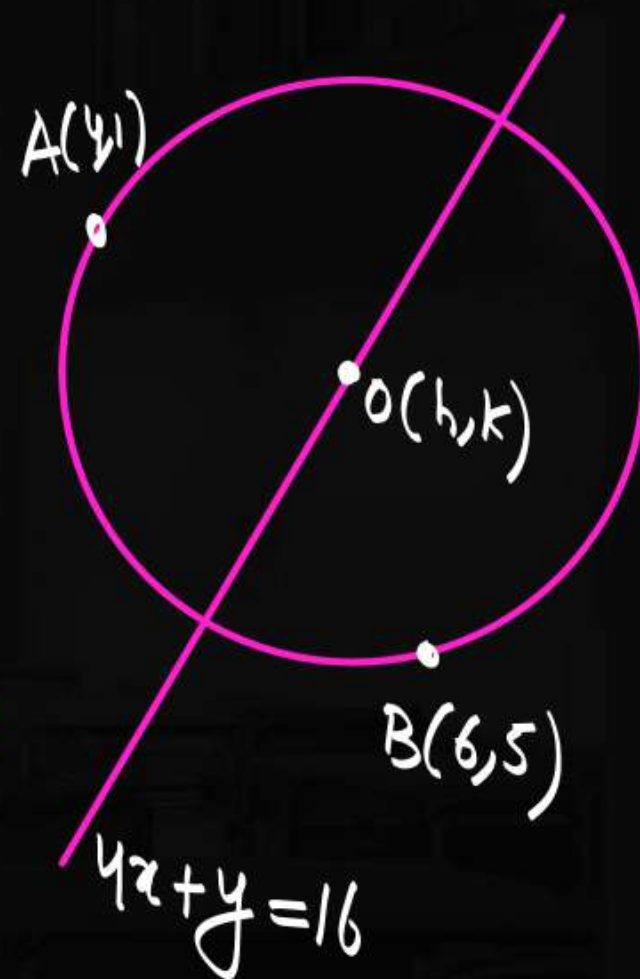
$$\Rightarrow \boxed{h+2k=11} \quad \text{--- (4)}$$

Since $O(h, k)$ lies on the line $4x + y = 16$

$$\text{So } 4h + k = 16 \Rightarrow \boxed{k = 16 - 4h} \quad \text{--- (5)}$$

Using (5) in (4), we get;

$$h + 2(16 - 4h) = 11$$



$$\Rightarrow h + 32 - 8h = 11.$$

$$\Rightarrow 7h = 21$$

$$\Rightarrow \boxed{h = 3}$$

So $k = 16 - 4 \times 3$

$$\Rightarrow \boxed{k = 4.}$$

Thus from ②, we get

$$r^2 = (4-3)^2 + (1-4)^2$$

$$\Rightarrow r^2 = 1 + 9$$

$$\Rightarrow \boxed{r^2 = 10.}$$

Hence from ①, we get;

$$(x-3)^2 + (y-4)^2 = 10.$$

$$\Rightarrow x^2 + 9 - 6x + y^2 + 16 - 8y = 10$$

$$\Rightarrow \boxed{x^2 + y^2 - 6x - 8y + 15 = 0.} \text{ which is the}$$

required equation of the circle.

QUESTION

Find the equation of the circle passing through the points $(2, 3)$ and $(-1, 1)$ and whose centre is on the line $x - 3y - 11 = 0$.

Homework.

QUESTION

Find the equation of the circle with radius 5 whose centre lies on x -axis and passes through the point $(2, 3)$.

(Sol) Suppose that the equation of circle be:

$$(x-h)^2 + y^2 = 25 \quad \text{--- (1)}$$

Since (1) passes through $(2, 3)$. So

$$(2-h)^2 + 3^2 = 25$$

$$\Rightarrow (2-h)^2 = 25 - 9$$

$$\Rightarrow (2-h)^2 = 16$$

$$\Rightarrow (2-h) = \pm \sqrt{16}$$

$$\Rightarrow 2-h = \pm 4$$

$$\Rightarrow 2-h = 4, -4$$

$$\Rightarrow h = -2, 6$$

Therefore from (1), the required circles are:

$$(x-6)^2 + y^2 = 25$$

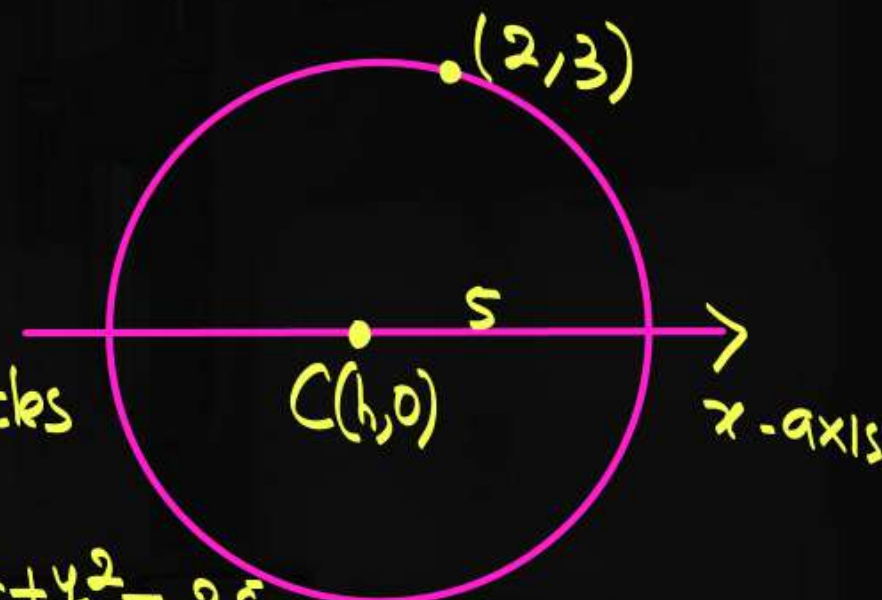
$$\Rightarrow x^2 + 36 - 12x + y^2 - 25 = 0$$

$$\Rightarrow x^2 + y^2 - 12x + 11 = 0$$

$$(x+2)^2 + y^2 = 25$$

$$\Rightarrow x^2 + 4 + 4x + y^2 - 25 = 0$$

$$\Rightarrow x^2 + y^2 + 4x - 21 = 0$$



QUESTION

Find the equation of the circle passing through $(0, 0)$ and making intercepts a and b on the coordinate axes.

(Sol) Clearly C divides the Diameter in $1:1$ ratio

So, By Mid-point formula

$$C(h, k) = C\left(\frac{a}{2}, \frac{b}{2}\right)$$

Also By Pythagoras theorem, we get;

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow (2r)^2 = a^2 + b^2$$

$$\Rightarrow 4r^2 = a^2 + b^2$$

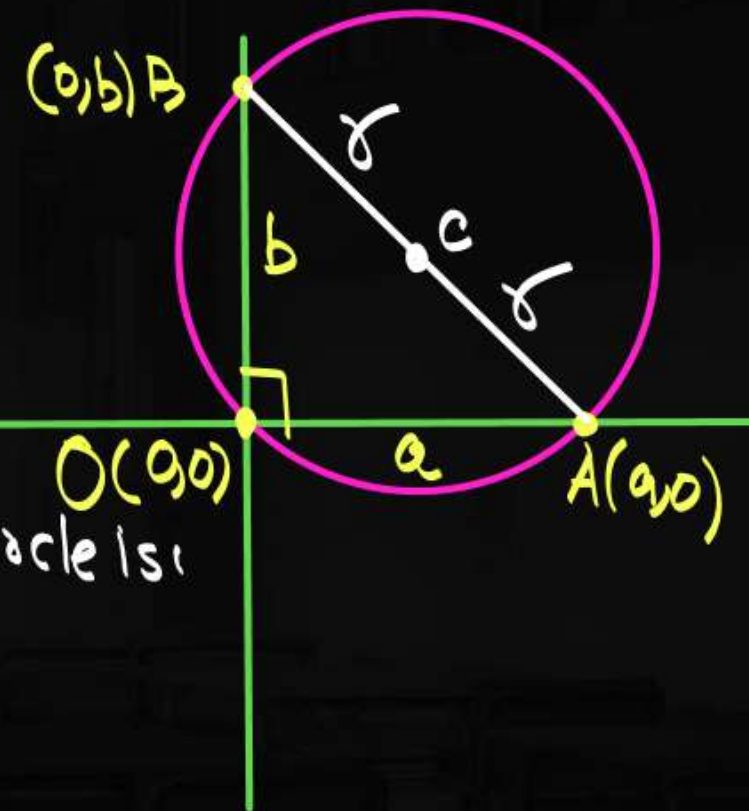
$$\Rightarrow r = \frac{1}{2} \sqrt{a^2 + b^2}$$

Therefore the required equation of the circle is:

$$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2 + b^2}{4}$$

$$\Rightarrow x^2 + \frac{a^2}{4} - ax + y^2 + \frac{b^2}{4} - by = \frac{a^2}{4} + \frac{b^2}{4}$$

$$\Rightarrow x^2 + y^2 - ax - by = 0$$



QUESTION

Find the equation of a circle with centre $(2, 2)$ and passes through the point $(4, 5)$.

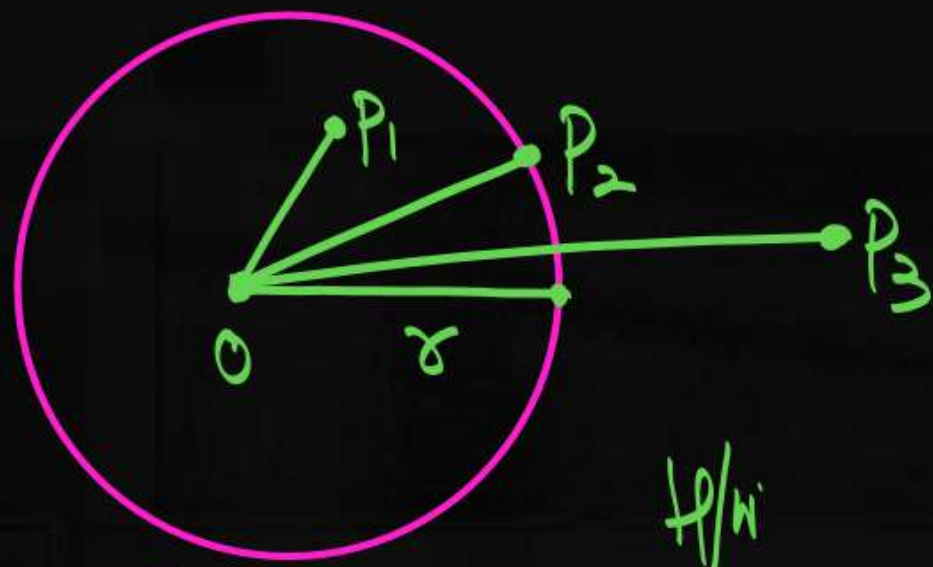
$$\underline{(x-2)^2 + (y-2)^2 = 8}$$

HP/W

QUESTION

Does the point $(-2.5, 3.5)$ lie inside, outside or on the circle $x^2 + y^2 = 25$?

$(0,0)$ $r = 5$



$OP_1 < r$

$OP_2 = r$

$OP_3 > r$

h/w

QUESTION

Two vertices of an equilateral triangle are $(-1, 0)$ and $(1, 0)$, and its third vertex lies **above x-axis**, then the **equation of the Circumcircle** of the triangle is :

Homework.

1 $x^2 + y^2 - \frac{2}{\sqrt{3}}x + \sqrt{3} = 0$

2 $x^2 + y^2 - \frac{2}{\sqrt{3}}y - \sqrt{3} = 0$

3 $x^2 + y^2 - \frac{2}{\sqrt{3}}y - 1 = 0$

4 $x^2 + y^2 - \frac{2}{\sqrt{3}}y + 1 = 0$

QUESTION(JEE-MAINS 2019)

If the area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq.units , then c is equal to:

(Next lecture)

1 13

2 20

3 -25

4 25

QUESTION(JEE-MAINS 2019)

A circle touching the **x-axis at $(3, 0)$** and making an intercept of length 8 on the y-axis passes through the point :

(Homework Problem)

- 1** $(3, 10)$
- 2** $(2, 3)$
- 3** $(1, 5)$
- 4** $(3, 5)$

Rasib Ayoub Rathed 

Syed Fakira 

Falak Altaf 

$$e^{2\pi} + 1 = 0.$$

Thank You